Architecture
PORTFOLIO

Ildea van den Berg | M.Arch.Tech | B.Tech

ILDEA VAN DEN BERG

My approach to design is rooted in curiosity and problem-solving, which drove my master's research on repurposing discarded local stone for contemporary architecture, through hands-on exploration and finding meaningful ways to connect traditional materials with modern practices. Compassion, endurance, and self-expression shape my work ethic and my commitment to thoughtful design.

Meticulous and dedicated, I take pride in approaching every task carefully, ensuring that each project reflects purpose, responsibility, and creativity.

Email: ildeavdb@gmail.com
Contact: 083 789 5713
www.linkedin.com/in/ildea-van-den-berg
SACAP reg CSAT67621033

EXPERIENCE

Richter and Associates Architects
Candidate Senior Architectural Technologist
2021 - Feb 2024

Attending on- and off-site meetings, drafting municipal submissions and technical construction drawings, and assisting with design development and project proposals for locations in Zimbabwe, Kenya, and the DRC.

Archactive Architects
Part Time Intern
2017 - 2018

Attending on- and off-site meetings, assisting with drawing revisions using Caddie and ArchiCad, and meeting with company representatives to explore new products for design development.

REFERENCES

Rupert Richter | Director
Richter and Associates Architects
rupert@r-aa.co.za
083 450 3646

Elise van Gass | Owner
Archactive Architects
elise@archactive.co.za
083 448 5512

EDUCATION

M.Arch Tech (Master of Architecture in Architectural Technology)
Tshwane University of Technology
2023 - 2024

B-Tech Architectural Professional Degree
Tshwane University of Technology
2015 - 2019

National Senior Certificate
Hoërskool Oosterlig
2013

SOFTWARE

ArchiCad
Revit
Adobe Photoshop
Adobe InDesign
Adobe Illustrator
Lumion
Sketchup
MS PowerPoint
MS Word
MS Excell

01

Masters Dissertation - Innovative Approaches to Architectural Stonework

02

Structure Narrative - Rissik Street

03

Spirit and Material Equipoise - The Artefact

04

Practice Work - Kisumu Jubilee Bridge

05

Practice Work - Fishbay Residence

06

Practice Work - Springvale Primary School

07

Practice Work - Vheneka Residential Estate

01

Masters Dissertation - Innovative Approaches to Architectural Stonework: Examining Pretoria's local stone resources and machining techniques

Research focus

Natural discarded stone in contemporary construction, highlighting the challenges and opportunities.

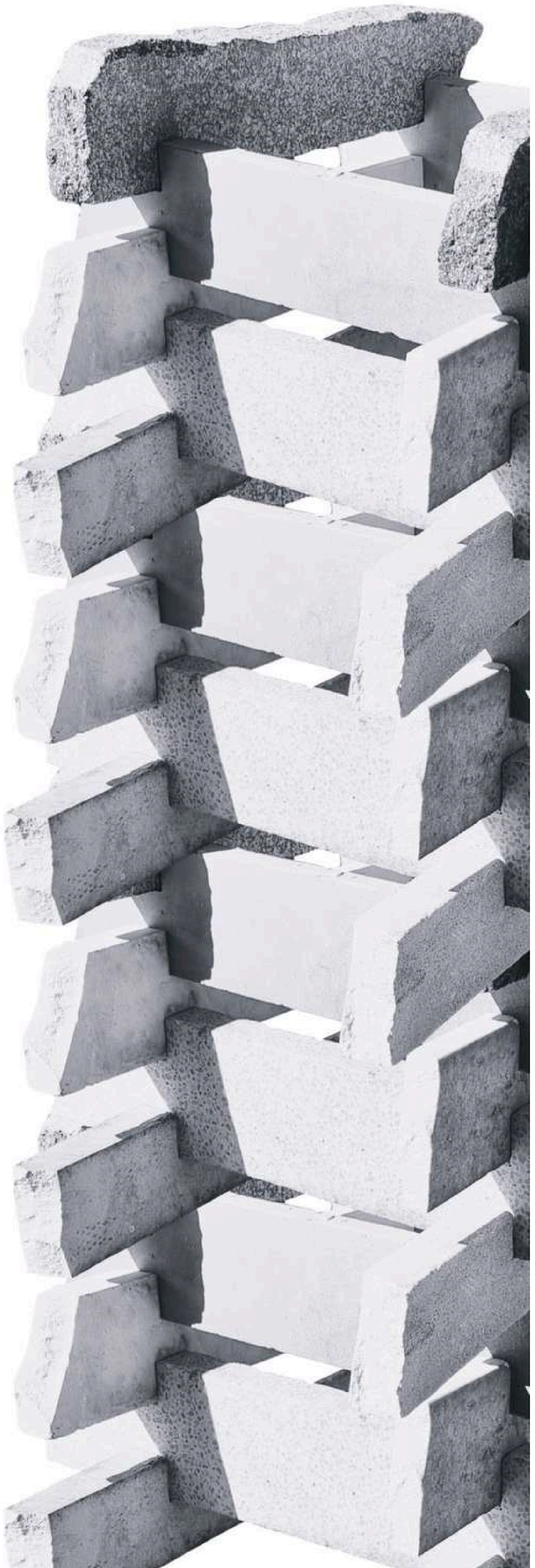
It covers the historical importance of the material, its uses today, and the potentiality for innovative reuse and technological improvements for applications in the future.

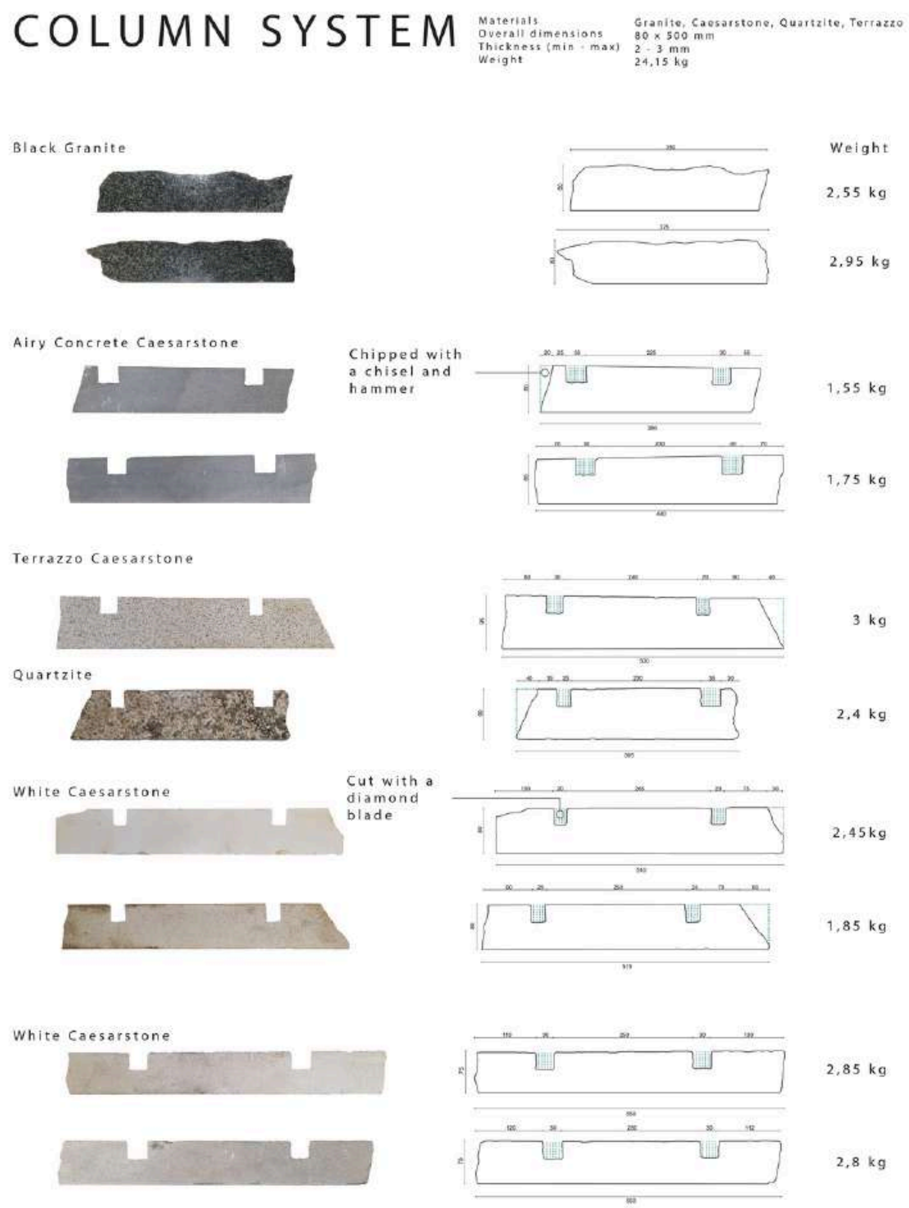
Column System: Repurposing Discarded Stone for Structural Use

The Column System explores the potential of repurposing discarded stone for construction by transforming offcuts into modular components. This approach presents a sustainable method for material reuse.

The process began with selecting and measuring stone pieces, cutting them into uniform 20 mm deep blocks while preserving their natural thickness. A grinder with a diamond blade created precise separation lines, followed by a hammer and chisel to refine the interlocking components for enhanced stability.

Once assembled, the system weighed 24.15 kg and was designed for easy disassembly and relocation. This project demonstrates how discarded stone can be reimagined as strong, reusable building elements, offering a practical solution for sustainable construction.





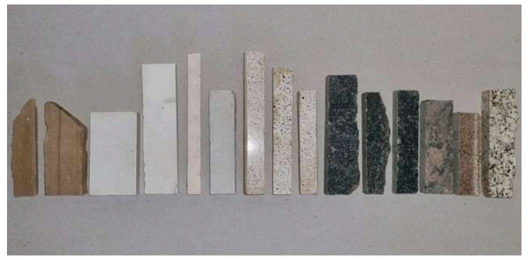
Column and Beam System: Reviving Discarded Stone for Structural Use

The Column and Beam System repurposes discarded stone by transforming it into a structural element. The process involved stacking and cutting stone samples to size using a grinder, then securing them with M6 screws, which were later cut into steel rods for reinforcement. A water-cooling system minimized dust and overheating during drilling, ensuring precise connections.

The final 3.64 kg assembly merges historical materiality with modern machining, demonstrating the enduring architectural potential of repurposed stone.

COLUMN & BEAM SYSTEM

Materials	Granite, Caesarstone, Quartzite, Terrazzo
Overall dimensions	6 x 220 mm
Thickness (min - max)	2 - 3 mm
Weight	3,64 kg



Stone Space Frame System: Reviving Discarded Stone

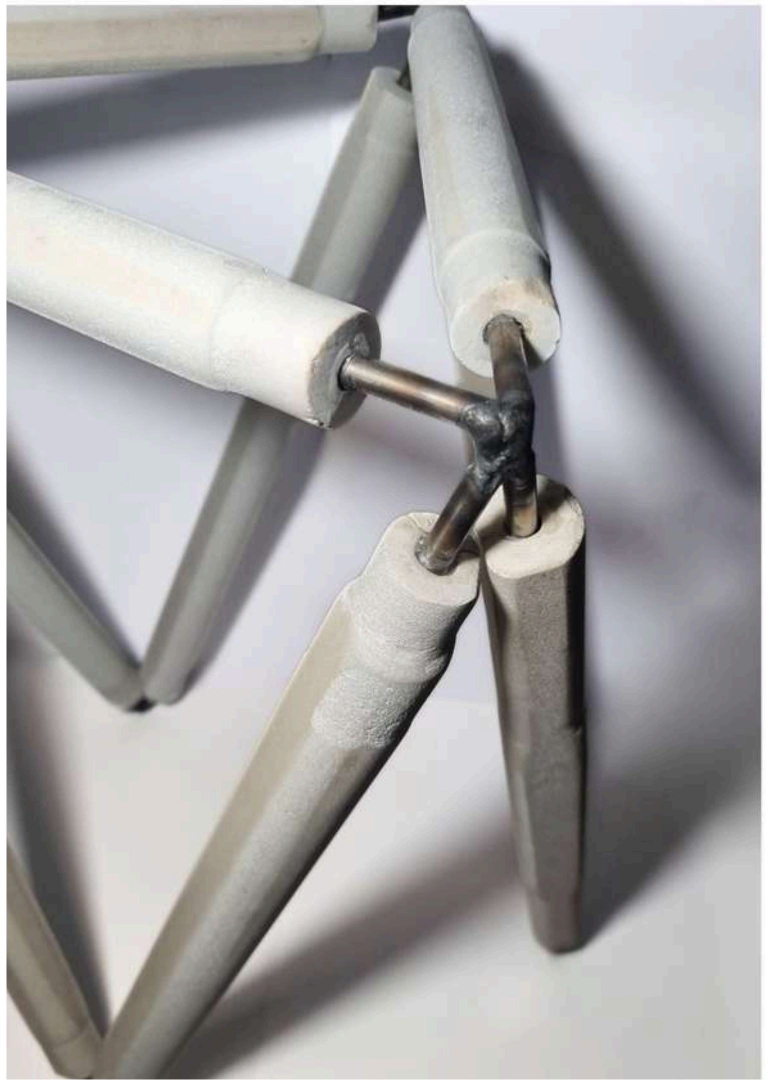
The Space Frame System was developed using discarded Caesarstone samples, selected for their uniform shape and texture to ensure smooth assembly. The pieces were cut to the required length using a grinder with a diamond blade, and a hose-dripping system, which prevented overheating, reduced dust, and minimised the risk of cracking.

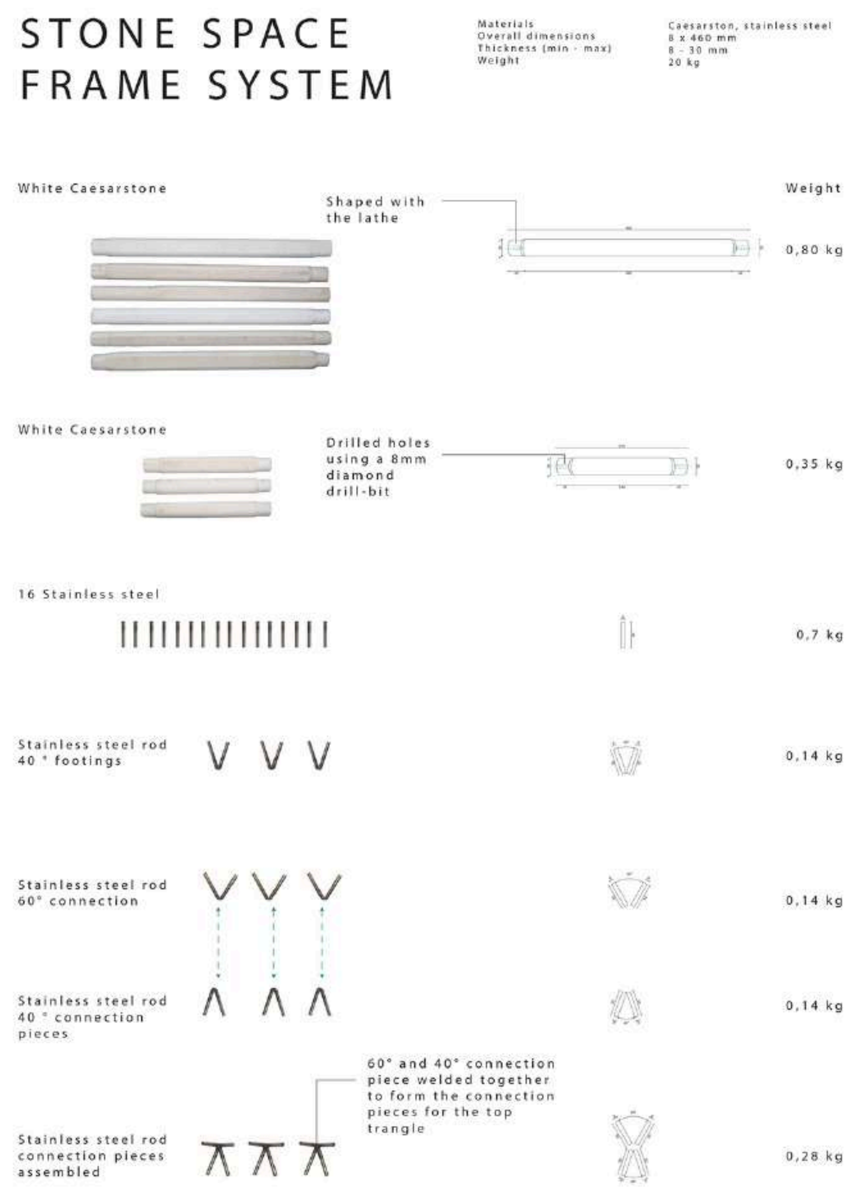
Each stone sample was drilled with 8 mm holes at its base using a diamond drill bit. However, during assembly, the square-shaped stones created alignment issues, as their sharp corners interfered with the steel connectors, preventing precise angles.

To resolve this, a local artisan skilled in metalwork and motorcycle repairs was consulted. His expertise was crucial in reshaping the blocks, eliminating obstructions, and allowing for accurate connections.

For structural stability, 8 mm stainless steel rods were chosen for their strength and compatibility with stone. To determine the best angles, flexible steel wire was shaped as a guide before the rods were welded into rigid connectors. The rods were then securely glued into the pre-drilled Caesarstone holes using epoxy, ensuring a strong and stable bond.

This system successfully demonstrates how discarded Caesarstone can be refined and repurposed for structural applications, bridging traditional material reuse with modern fabrication techniques.





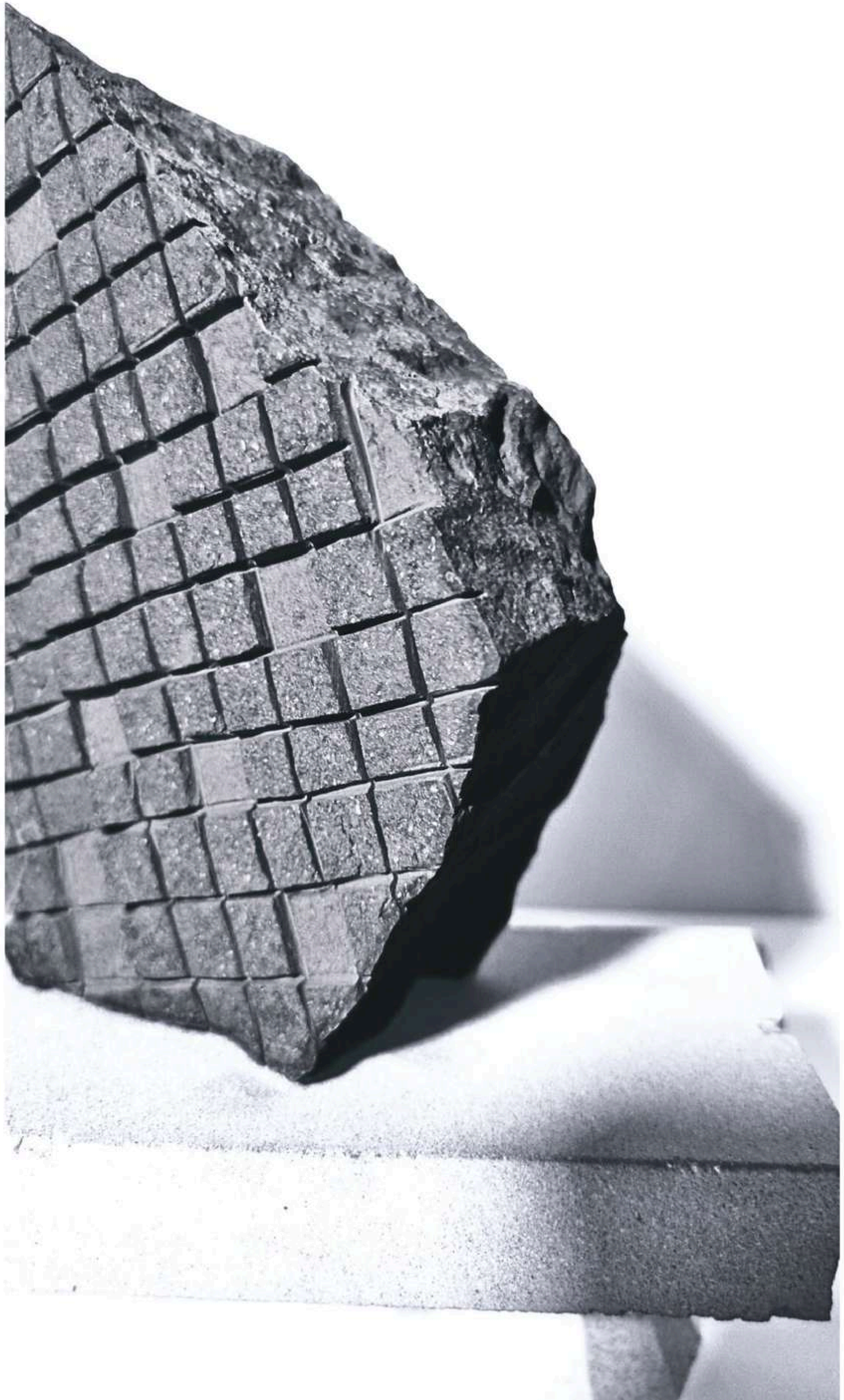







Through testing various stone types and machining techniques—including grinding, drilling, chipping, polishing, and engraving—the distinct properties of each material became evident. These tests provided insights into how different stones respond to manipulation, highlighting their potential applications in architecture.

By understanding how each stone reacts to machining, architects can make more informed material choices, aligning their selections with design intent. These findings serve as a reference for integrating natural stone into contemporary architecture in ways that respect its inherent characteristics while expanding its possibilities in modern construction.



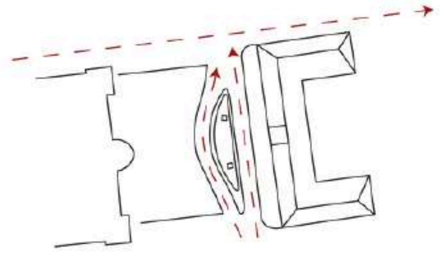
02

5th year Masters project
Structure Narrative - Rissik Street

Project Overview

This project explores the relationship between urban decay, historical significance, and the human experience within architectural spaces.

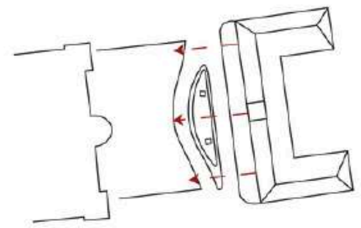
The site is located between historical structures and a fountain, the project reimagines the role of architecture in reconnecting people to their environment and history



Site access

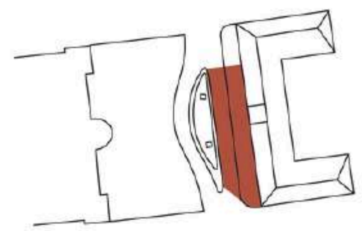
Site access from Helen Joseph street and market street.

Rissik street proposed to be temporarily closed for transportation to provide safe pedestrian access.



Views

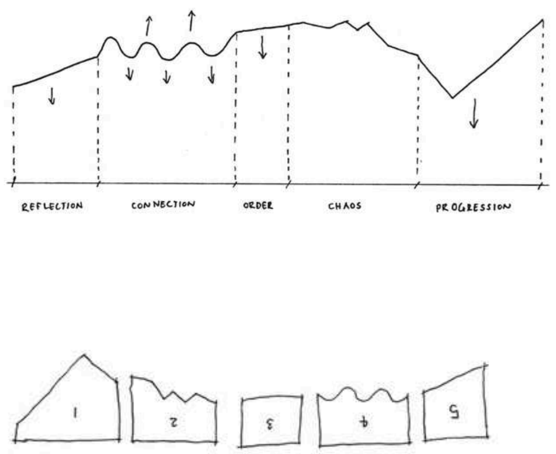
Important views from within the building looking out onto the fountain as well as looking up towards the building.

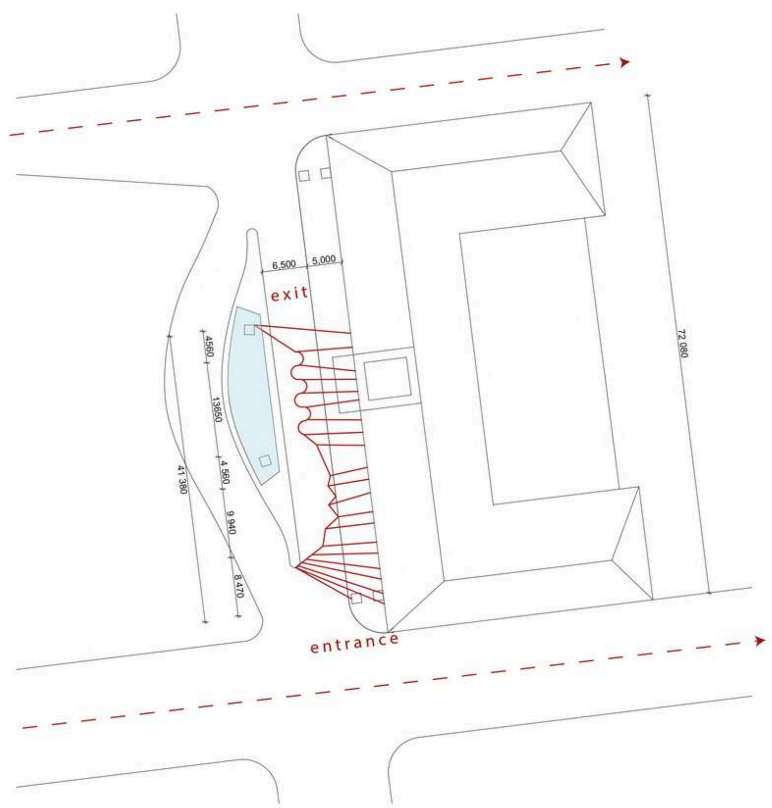


Proposed site location

Site to be located between the old building and the fountain.

the shape is designed according to my methodology and the history timeline.



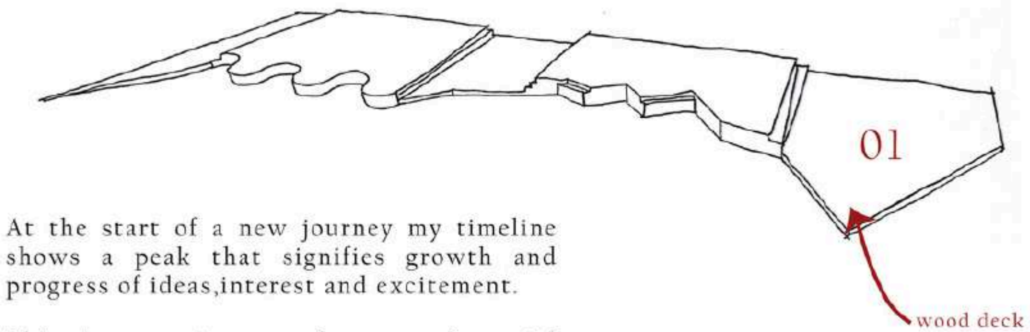


Design

five stages

progression

Graphical representation of my process and growth timeline:



At the start of a new journey my timeline shows a peak that signifies growth and progress of ideas,interest and excitement.

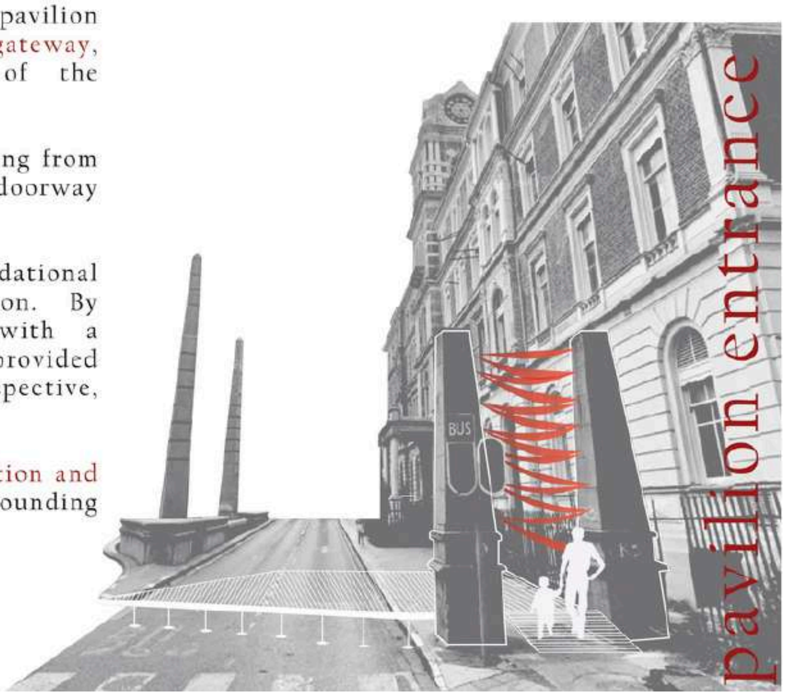
This "progress" stage has a substantial impact on how I process information and engage with my work.

The entry point of the pavilion serves as a symbolic gateway, representing the start of the journey.

It features red fabric hanging from existing pillars, creating a doorway for people to enter.

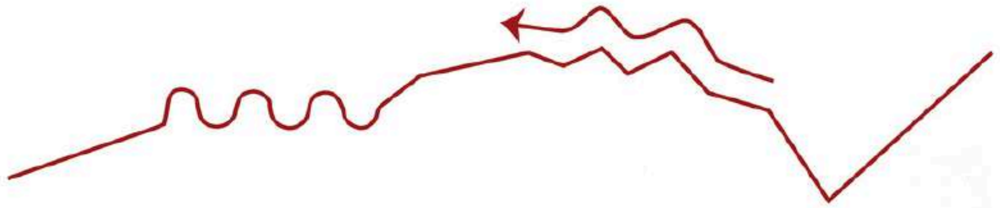
A wooden deck is the foundational platform for the pavilion. By elevating the pavilion with a wooden deck, visitors are provided with an elevated perspective, slightly above street level.

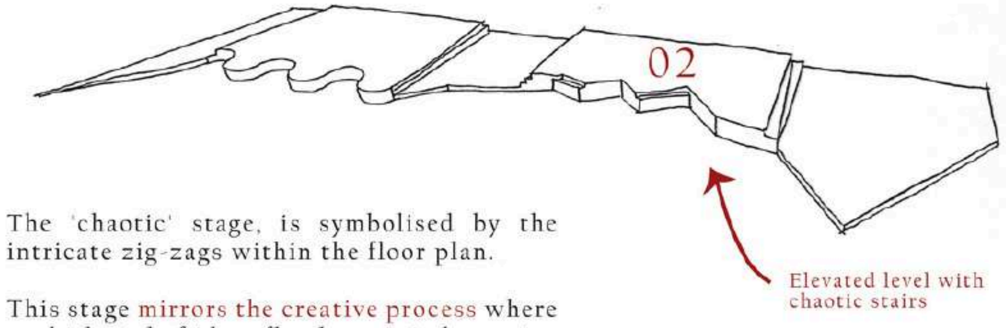
Creating a sense of connection and prominence within the surrounding environment.



Chaos

Graphical representation of my process and growth timeline:



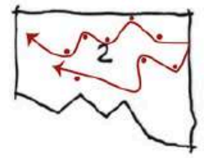


The 'chaotic' stage, is symbolised by the intricate zig-zags within the floor plan.

This stage mirrors the creative process where a whirlwind of ideas floods my mind, causing a overwhelming sense of chaos.

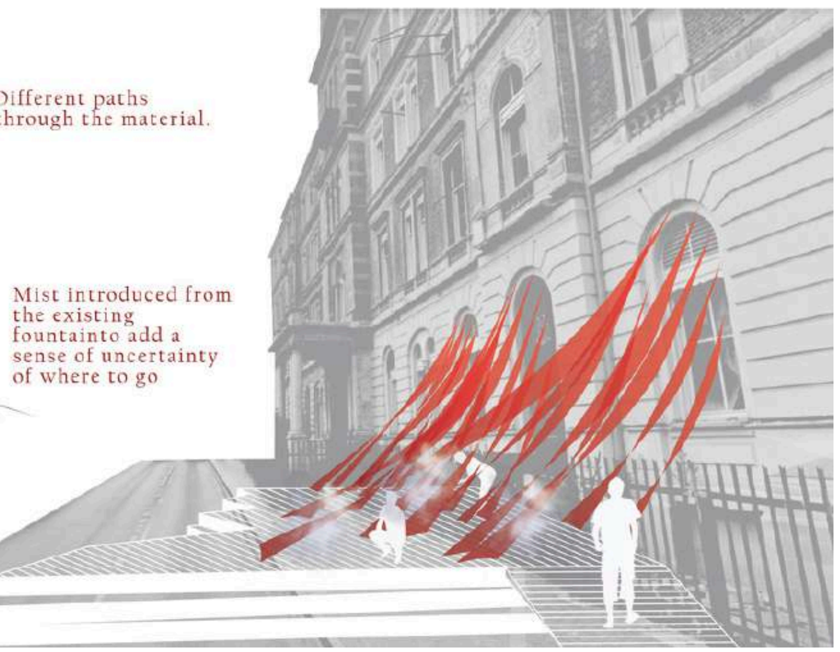
The zig-zags represent the constant ups and downs I navigate as I grapple with the abundance of ideas, contemplating how to incorporate them into the project or make the difficult decision to leave some behind.

This stage shows the beautiful chaos of creativity, where I weave through concepts and possibilities, ultimately shaping the unique and dynamic character of the pavilion.



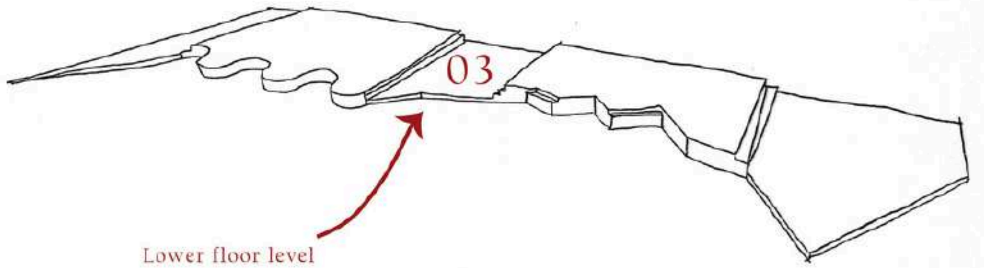
Different paths through the material.

Mist introduced from the existing fountainto add a sense of uncertainty of where to go



Order

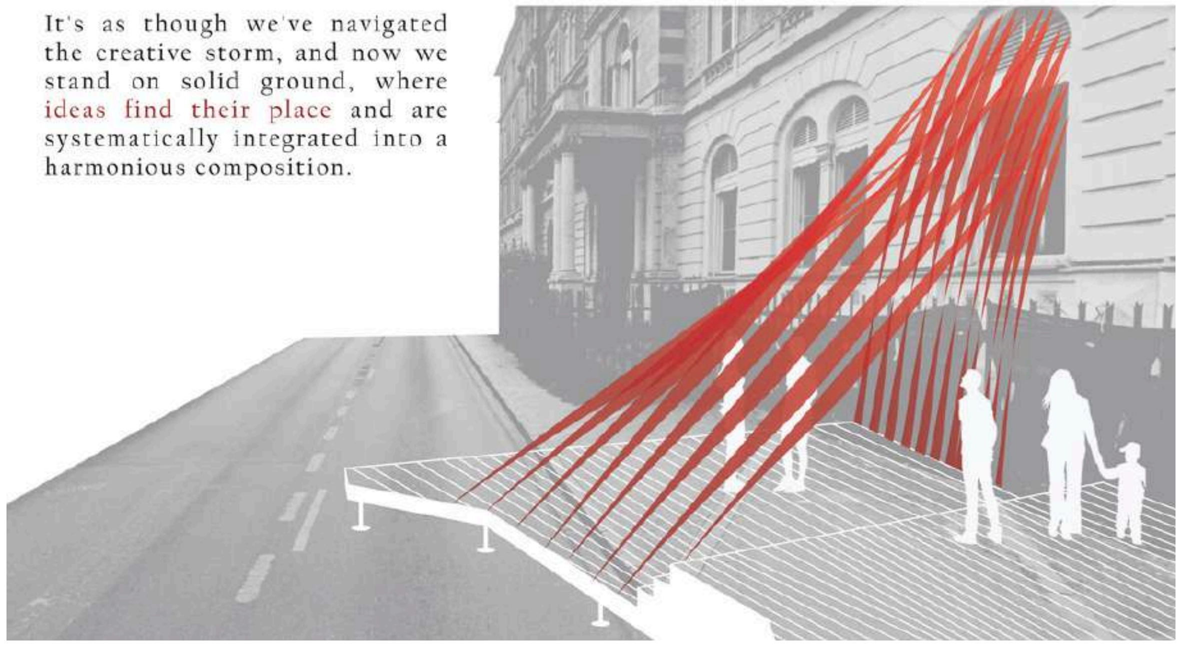
Graphical representation of my process and growth timeline:



The 'order' stage of the pavilion is a contrast to the chaotic stage. The platform descends, signifying a return to a sense of equilibrium and structure.

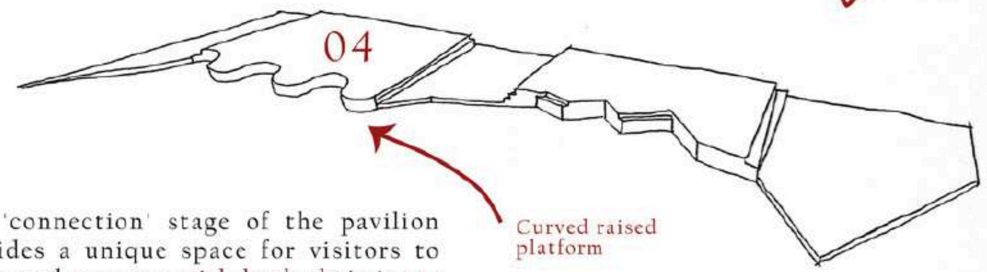
The carefully arranged material hanging from the old building reflects a new level of organization and order.

It's as though we've navigated the creative storm, and now we stand on solid ground, where ideas find their place and are systematically integrated into a harmonious composition.



Connection

Graphical representation of my process and growth timeline:

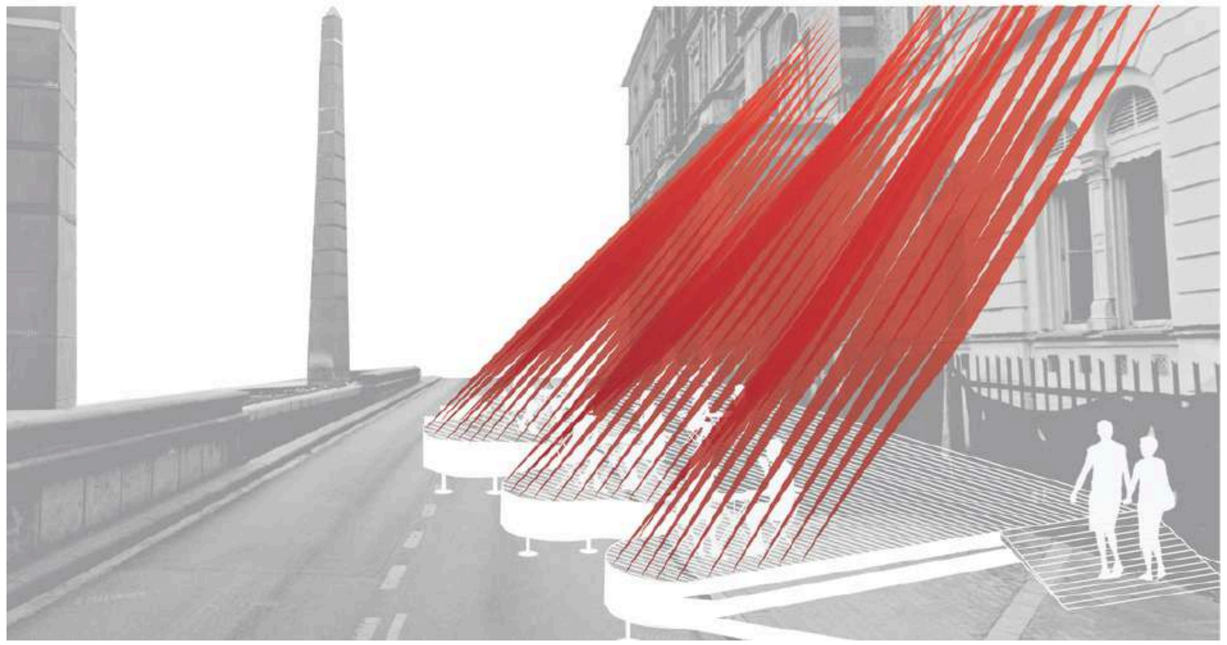


The 'connection' stage of the pavilion provides a unique space for visitors to pause and connect with both their inner thoughts and the surrounding environment.

The curved raised platform serves as a haven for those who have journeyed through the previous stages. The seating on this platform encourages people to connect with their emotions, thoughts, and the symbolism of the space.

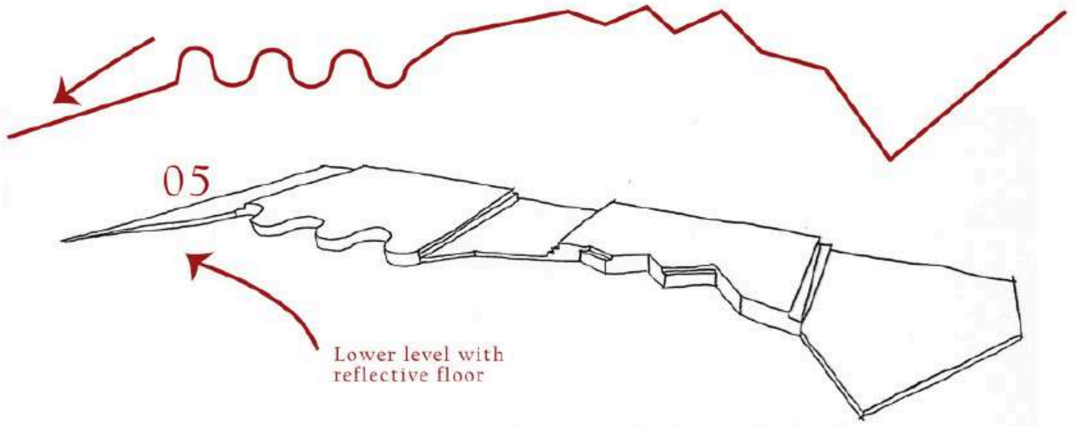
Much like my own creative journey, before reaching the end, visitors at this stage tend to look back at what they've done so far and what remains to be accomplished.

The presence of the fountain adds an auditory element to the experience:
The sound of flowing water is soothing, though its view remains partially obscured by the material draping from above.



Reflection

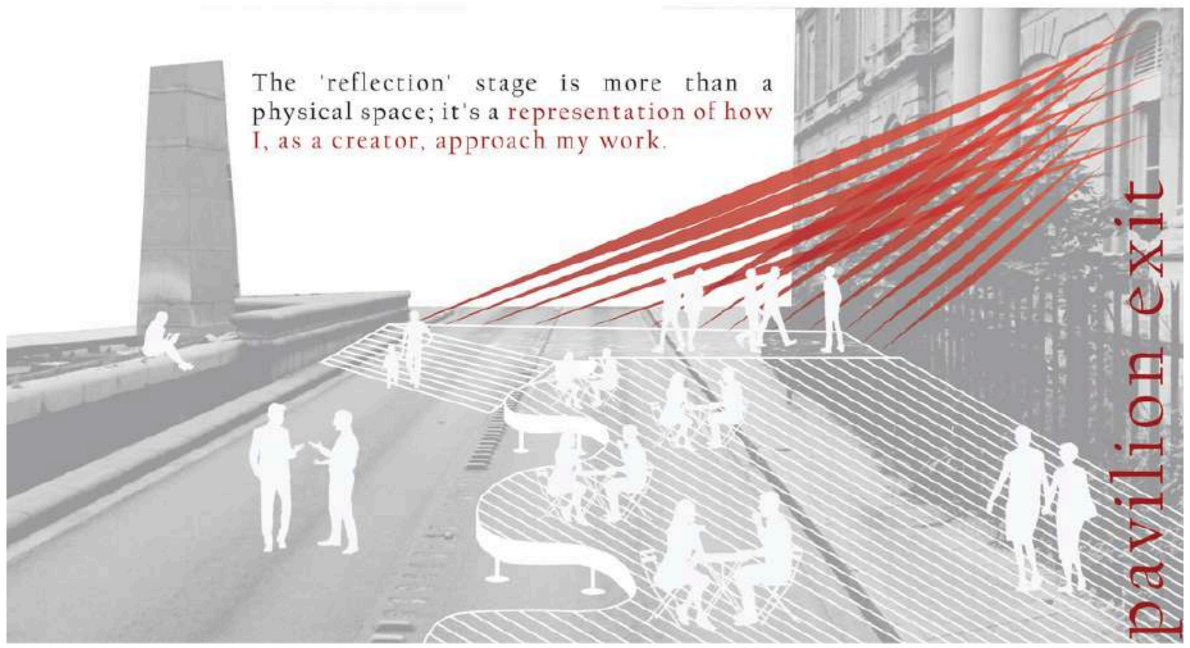
Graphical representation of my process and growth timeline:

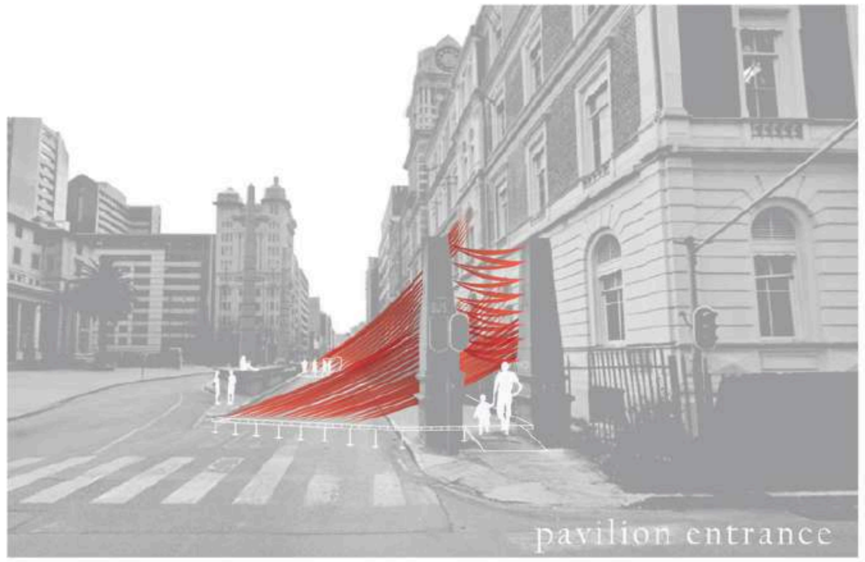


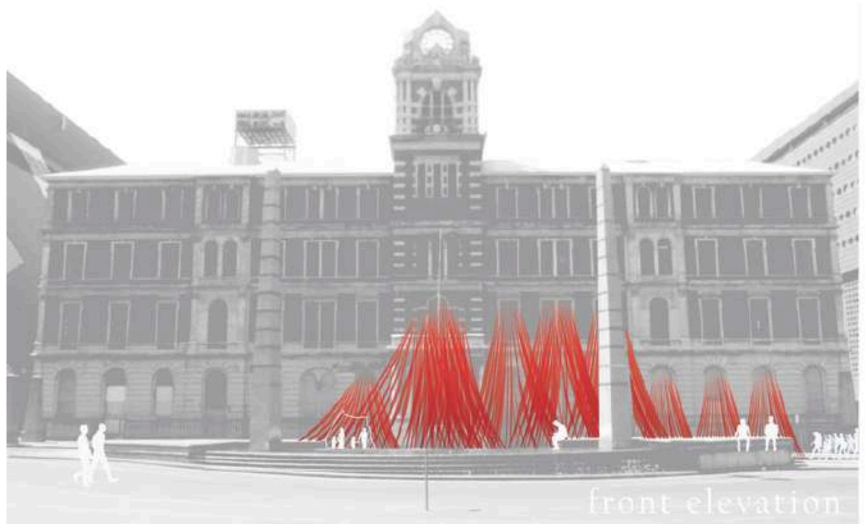
The 'reflection' stage is a space designed for contemplation and self-discovery. Here, the floor surface itself becomes a reflective canvas, symbolizing the opportunity for visitors to introspect and connect with the entirety of their journey.

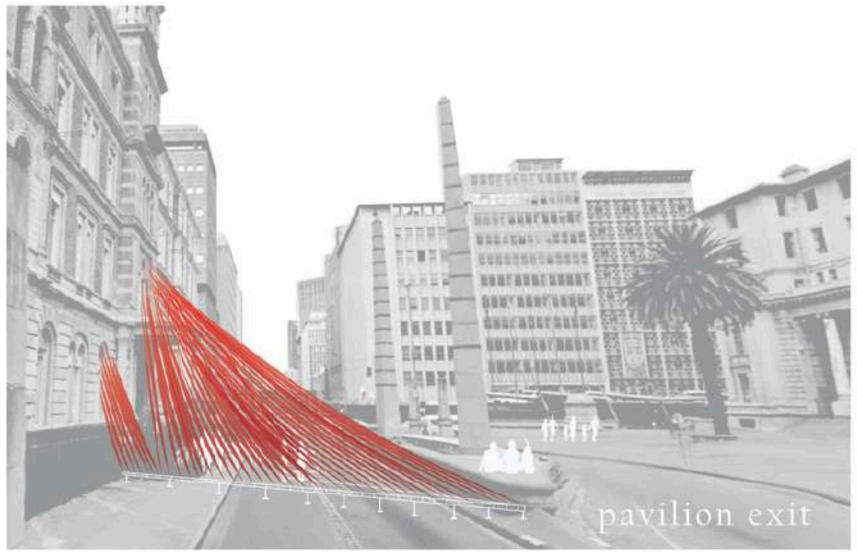
As visitors step onto the reflective surface, they are met with the gentle sounds of the nearby fountain, creating a soothing backdrop to their reflective experience. The reflective surface, mirroring the past and the stages they have traversed, invites individuals to metaphorically look back on their creative and personal voyage.

The path that leads out to the fountain is not only an invitation to connect with the water's serene presence but also an encouragement to reflect on the progression, chaos, order, connection, and personal growth that have been experienced.









Journey of reflection

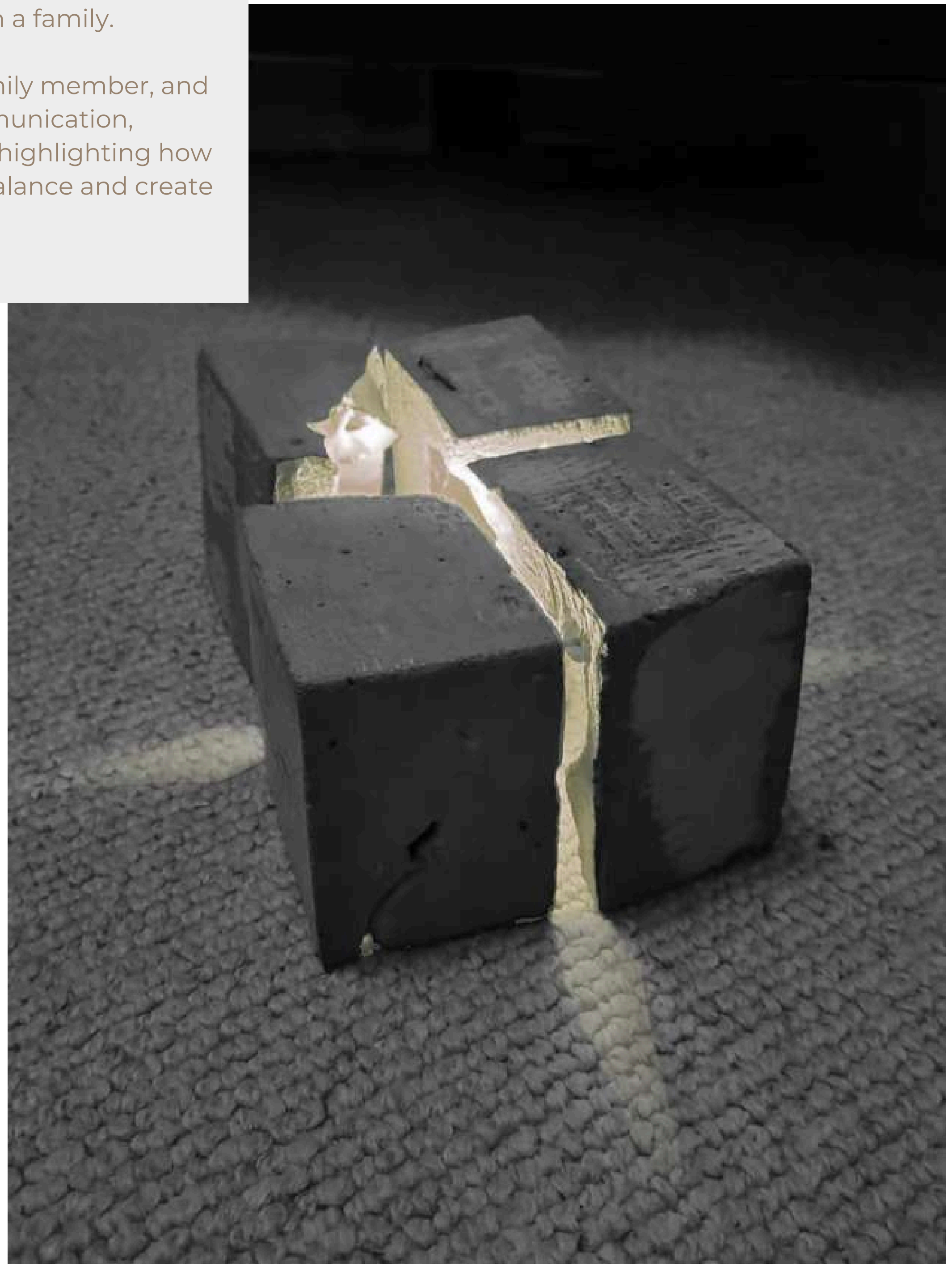
03

5th year Masters project
Spirit and Material Equipoise - The Artefact

Project Overview

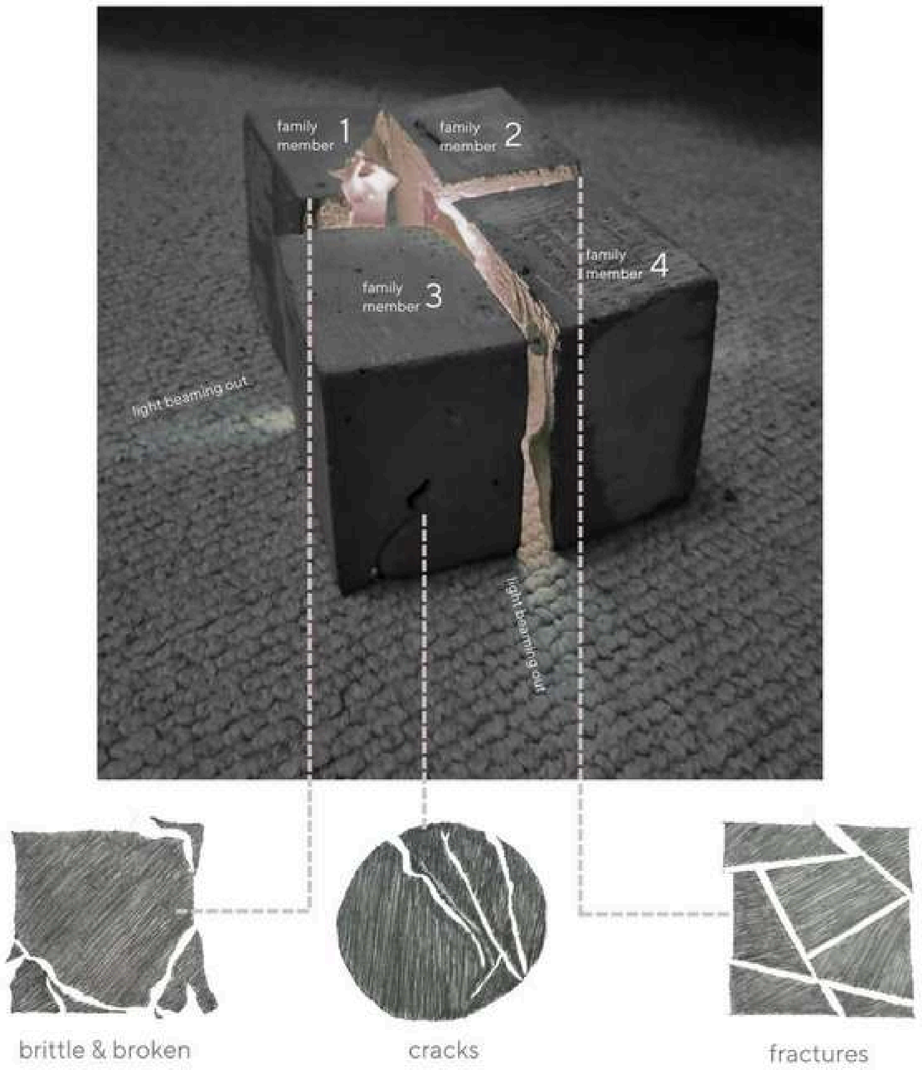
This project is about a symbolic dinner table made from Fastcrete to represent the complex dynamics within a family.

Each block signifies a family member, and explores themes of communication, connection, and conflict, highlighting how family tensions disrupt balance and create lasting impacts.

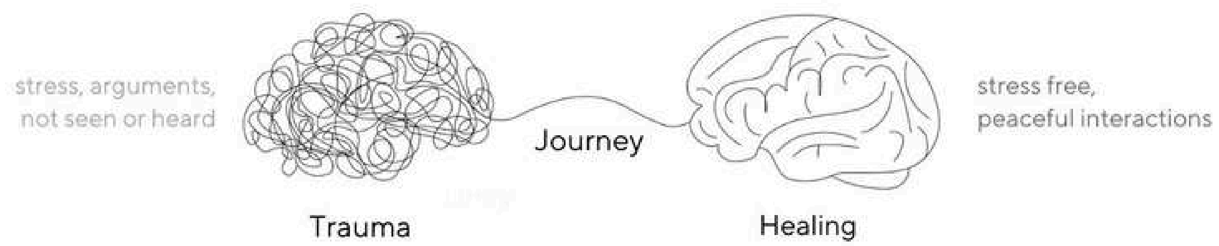


THE DINNER *Table*

The most contentious site in the home.
- a family of four -



The Journey from Trauma to Healing



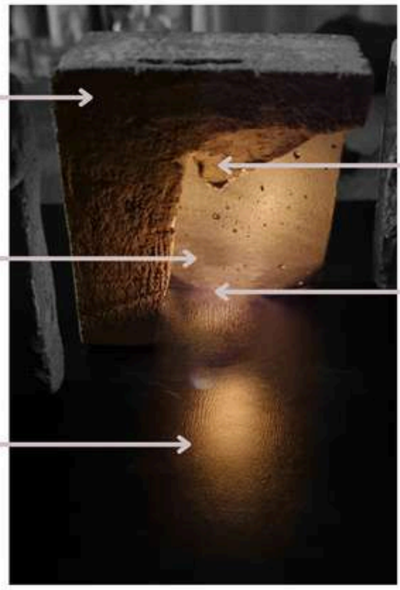
LIGHT & *Reflection*

The light illuminates beneath my section of the table, symbolizing the feeling of being unheard or unseen, serving as a signal for others in the family to acknowledge my presence.

smooth edges

light within

reflection



cracks

hidden fractures

Unity

Division

Sequence of fracture - Top View

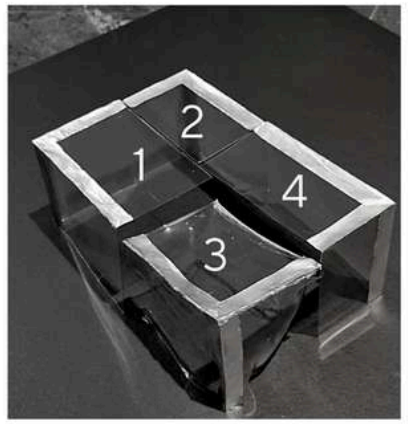
My father represents the largest block, having the most significant influence on our family dynamics. However, his presence leads to more fractures than the other blocks, highlighting the fact that while he has a substantial impact on us, he also causes internal turmoil within himself.

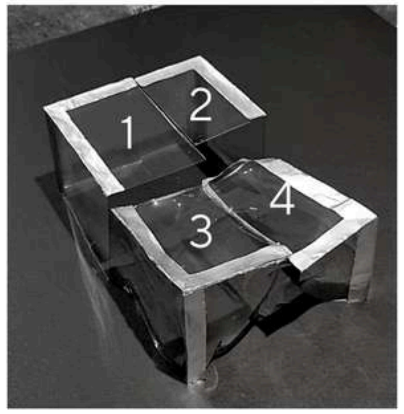
Within the family, I have been perceived as the perfectionist, which is why my block appears smooth on the outside. However, beneath its seemingly flawless surface, my block reveals deep fractures. The fractures shows that even though I strive for perfection and present a polished exterior, there are unresolved complexities that lie beneath.

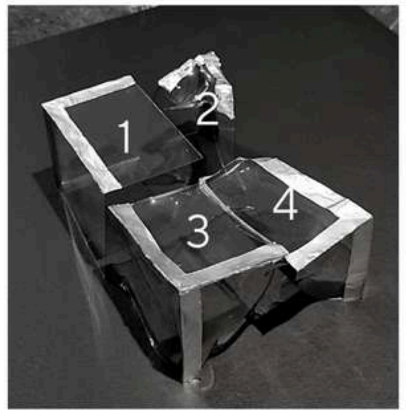
My mother represents the second largest block, characterized by imperfections along its edges. She is the person who strives to repair and heal the family. However, despite her efforts, her block displays visible cracks on the outside, symbolizing the challenges she faced.

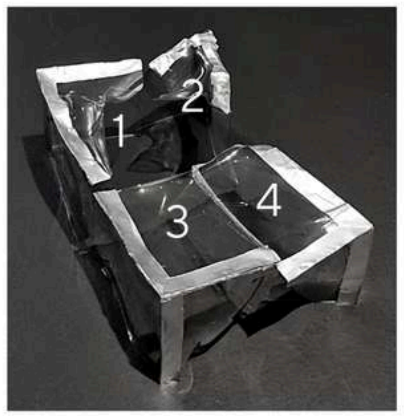
My sister embodies the final block.The thicker edges symbolize the barriers she erects to protect herself. Her block bears distinct, thicker marks inside, representing not only the unique challenges and struggles she carries within her, but also the resilience and strength that has developed as a result.

TABLE documentation









Outside factors and tension between the family members cause the table to show fractures.

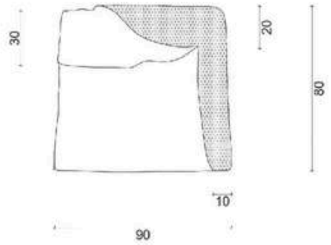
Over time the blocks representing the individuals start to show damage. Block 1 & 2 show the most trauma, because of the tension caused between them.

BLOCK 1

Material	Fastcrete, and metallic spray paint
Overall dimensions	90 x 80mm
Thickness (min - max)	10 - 20mm
Weight	317g

Inside View

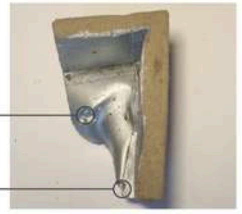
The inside of the block is painted with metallic paint to reflect the light source that radiates from block 2.



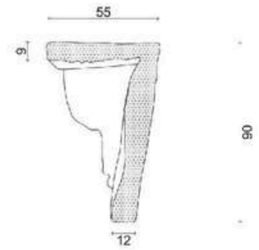
Bottom View

Fractures

Damage



Fractures and damage caused to the inside because of tension and trauma

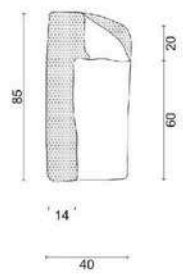


Front View

The bent ridges and distorted top visually represent the tension. The warped ridges symbolize the strain and pressure, showing visible signs of damage. The heat that caused the warping indicates the intensity of emotions and conflicts.

Damage

Fractures

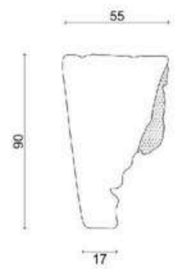


Top View

Fractures

Seperation Ridge

The block's exterior boasts a smooth finish, concealing its contents within.



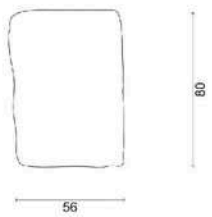
Back View

The bent ridges and distorted top visually represent the tension.

The warped ridges symbolize the strain and pressure, showing visible signs of damage.

Fractures

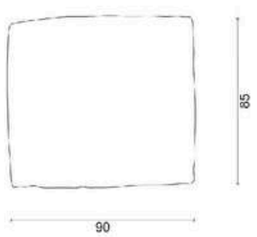
Seperation Ridge



Side View

The heat that caused the warping indicates the intensity of emotions and conflicts.

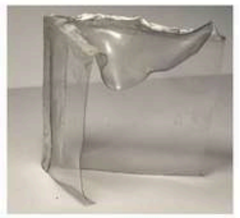
Seperation Ridge



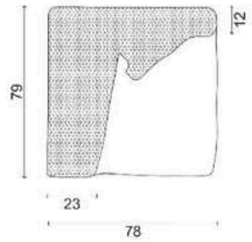
BLOCK 2

Material	Fastcrete, and metallic spray paint
Overall dimensions	78 x 79mm
Thickness (min - max)	10 - 23mm
Weight	302 g

Inside View



The inside of the block is painted with metallic paint to reflect the light source that radiates from block 2.

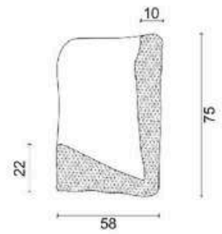


Bottom View

Fractures

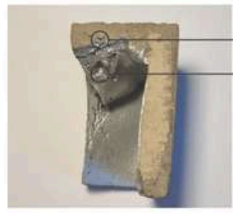
Damage

The fractures and damage is significantly more than the rest of the blocks.



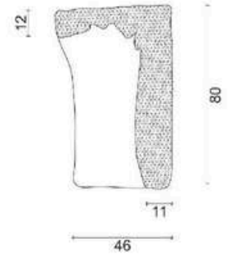
Front View

The sides of the blocks curve inward, as if someone sought to turn inward and separate themselves from the rest of the blocks.



Fractures

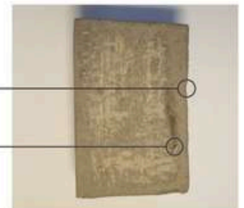
Damage



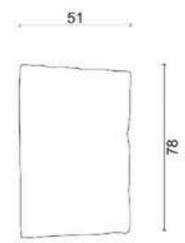
Top View

Seperation Ridge

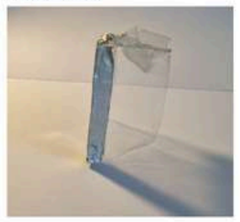
Fractures



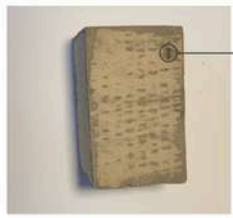
The block's exterior boasts a smooth finish, but fractures start to show on the outside.



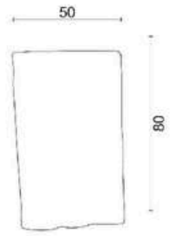
Back View



The back and side views of the blocks appear less bent compared to the top and front views, indicating that the trauma is primarily influenced because of block 1.

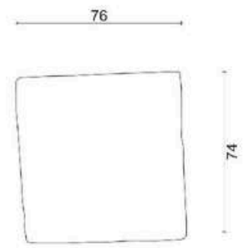


Fracture



Side View

Seperation Ridge



BLOCK 3

Material
Overall dimensions
Thickness (min - max)
Weight

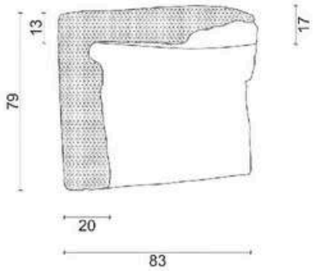
Fastcrete, and metallic spray paint
85 x 77 mm
10 - 25mm
317g

Inside View

The inside of the block is painted with metallic paint to reflect the light source that radiates from block 2.

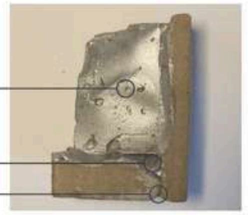
Damage and fractures are visible.

Damage
Fractures
Fractures

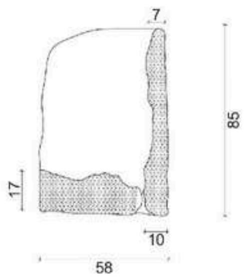


Bottom View

Damage
Fractures
Cracks



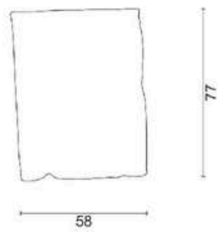
The two ridges have split from one another revealing a crack and fractures inside.



Front View

The outside of the blocks appears smooth, except for a visible crack that runs through the front view, resulting from the separation between the two ridges.

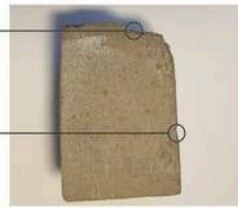
Cracks
Fractures



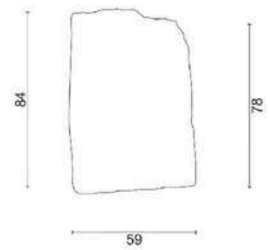
Top View

Seperation Ridge

Seperation Ridge



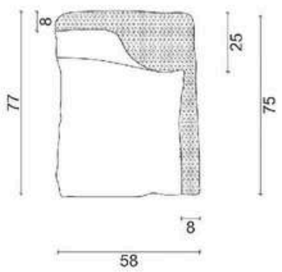
The block's exterior boasts a smooth finish, concealing its contents within.



Back View

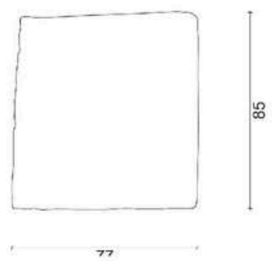
The bent ridges and distorted top visually represent the tension.

Fractures



Side View

Seperation Ridge



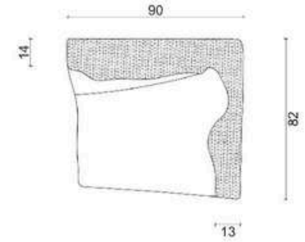
BLOCK 4

Material
Overall dimensions
Thickness (min - max)
Weight

Fastcrete, and metallic spray paint
80 x 81mm
10 - 12mm
372 g

Inside View

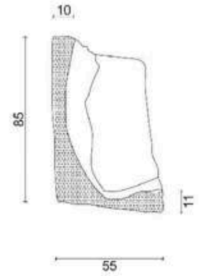
The inside of the block is painted with metallic paint to reflect the light source that radiates from block 2.



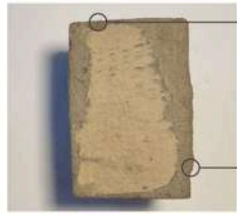
Bottom View

Fractures
Damage

Fractures and damage is visible in the deformaty of the ridges and inside.

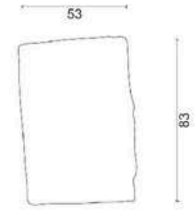


Front View



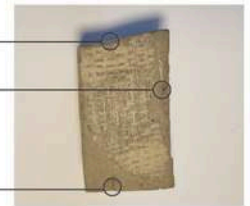
Seperation Ridge

Seperation Ridge

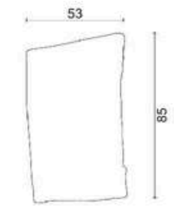


Top View

Seperation Ridge
Fractures
Cracks



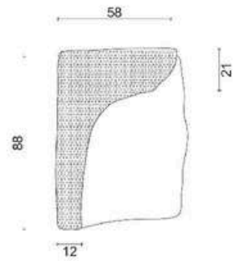
The block's exterior boasts a smooth finish, concealing its contents within.



Back View

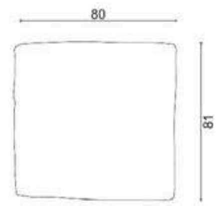
The sizes of the ridges on the block vary, creating an imbalance and making it slightly top-heavy, symbolizing the presence of heavy thoughts within.

Fractures
Seperation Ridge



Side View

Seperation Ridge



04

Practice Work
Kisumu Jubilee Bridge

Project Overview

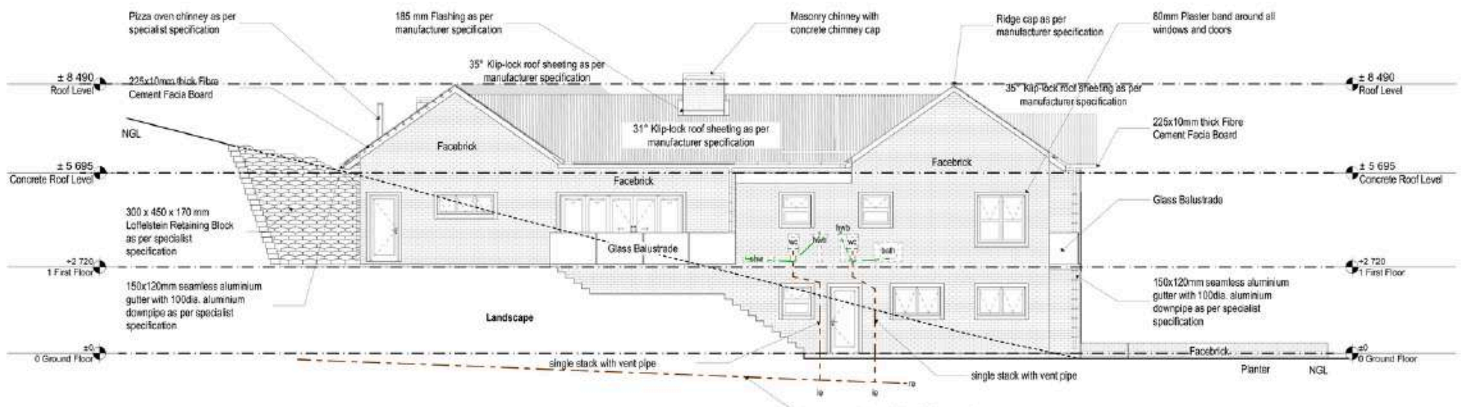
The Kisumu Market Regeneration Project is a transformative initiative aimed at upgrading the existing informal traders' market in Kisumu, Kenya.

Originally intended to provide a new trading platform for local vendors, the project evolved into a comprehensive urban development plan.

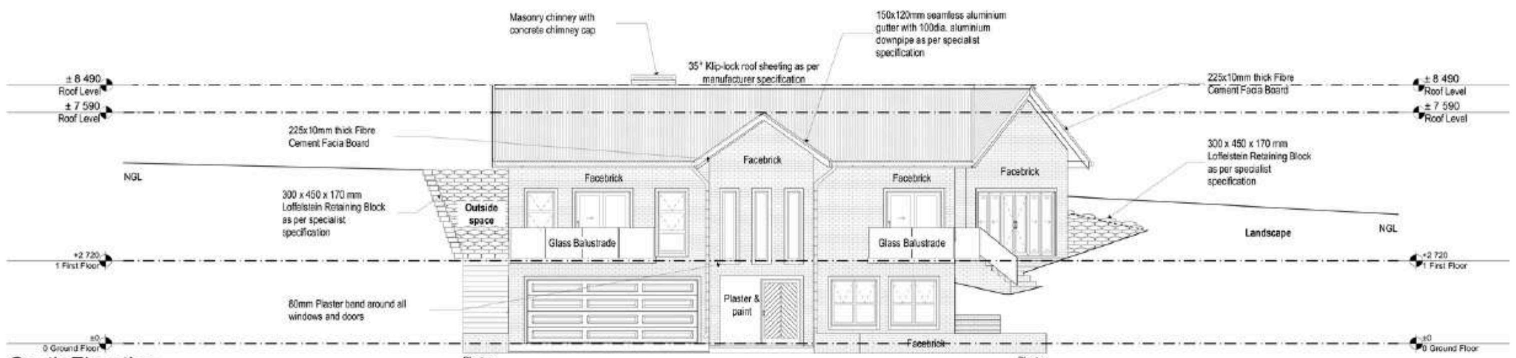
Site Plan

05

Practice Work
Fishbay Residence



West Elevation
Scale 1:100

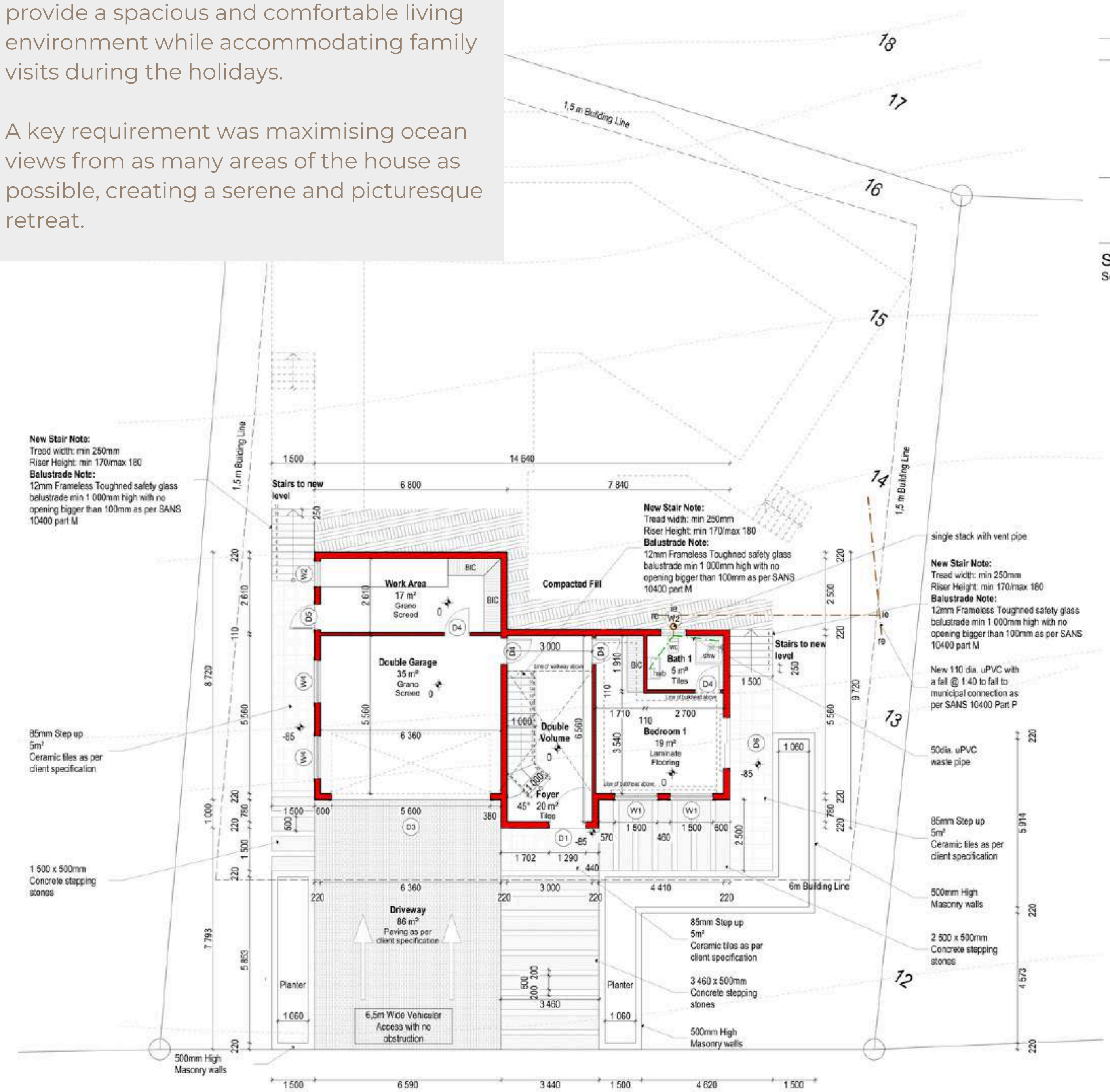


South Elevation
Scale 1:100

Project Overview

This retirement home was designed to provide a spacious and comfortable living environment while accommodating family visits during the holidays.

A key requirement was maximising ocean views from as many areas of the house as possible, creating a serene and picturesque retreat.



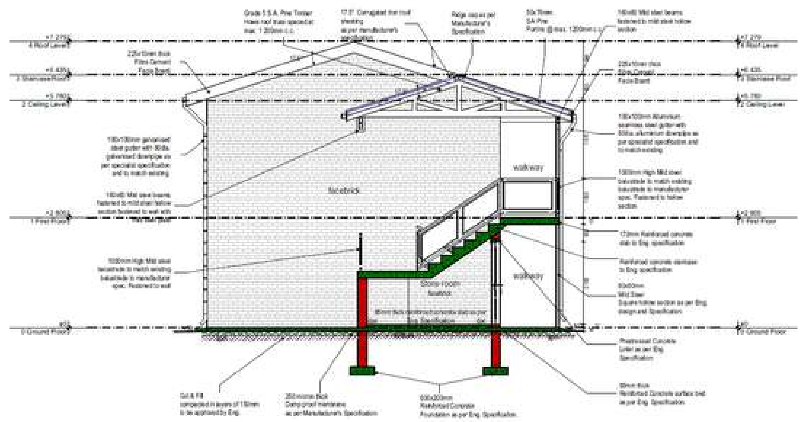
06

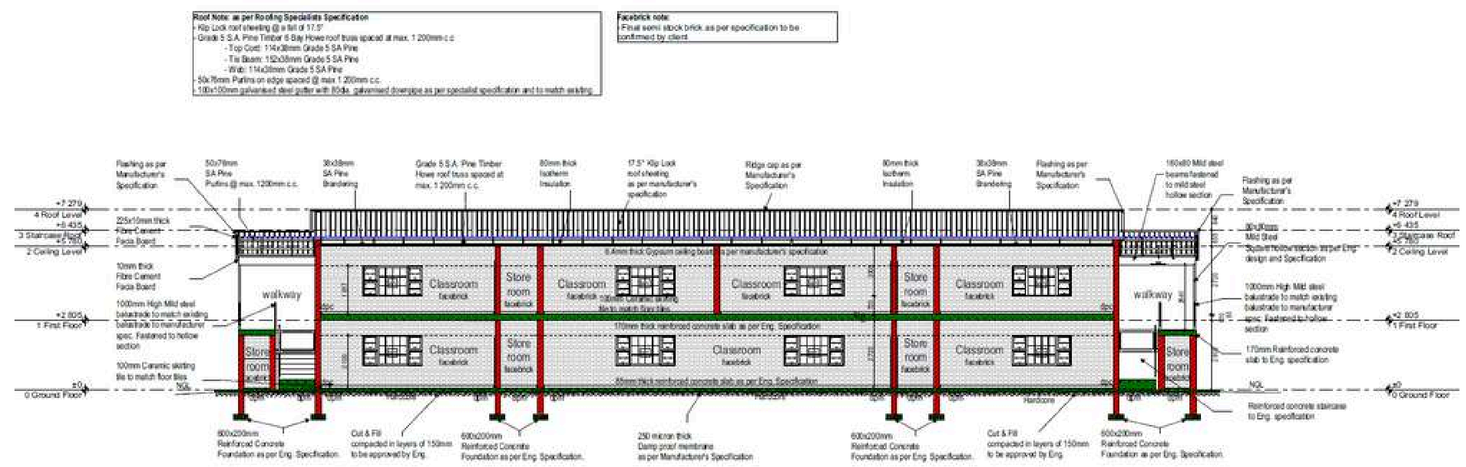
Practice Work
Springvale Primary School

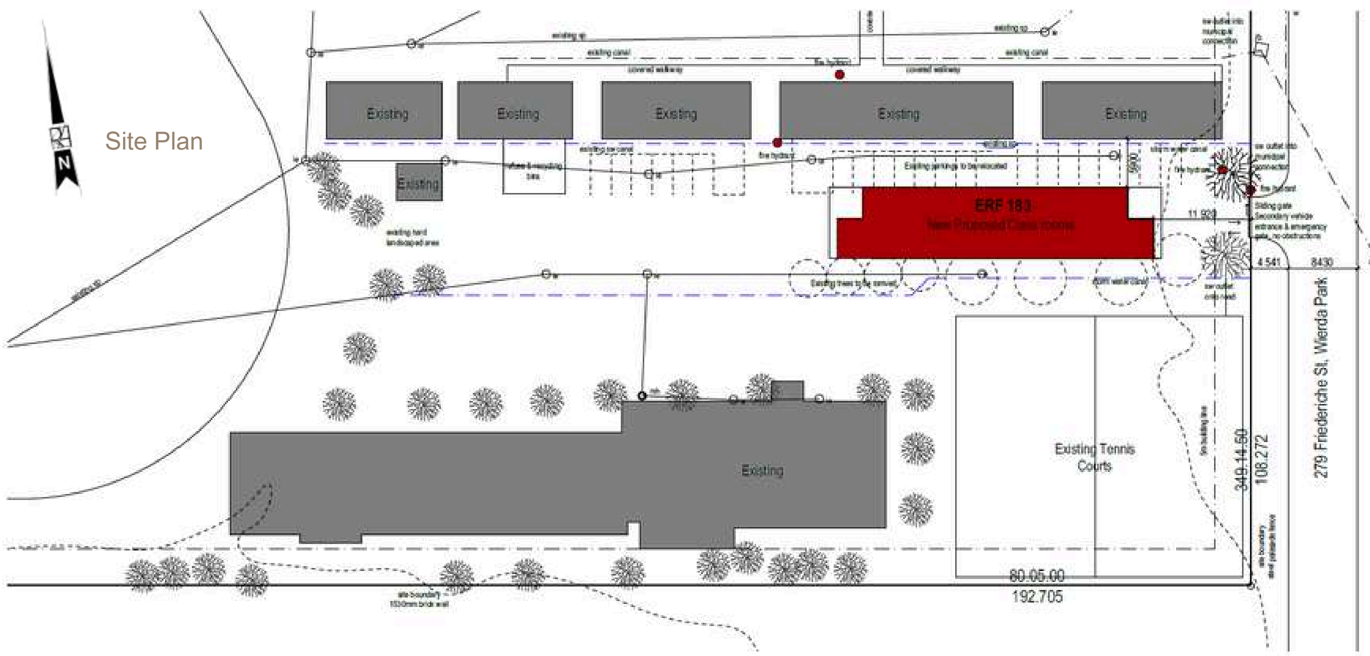
Project Overview

The Springvale Primary School expansion project involved designing and constructing additional classrooms to accommodate increasing student enrollment.

A key requirement was ensuring that the new classrooms seamlessly integrated with the existing school buildings in terms of materials and aesthetics.







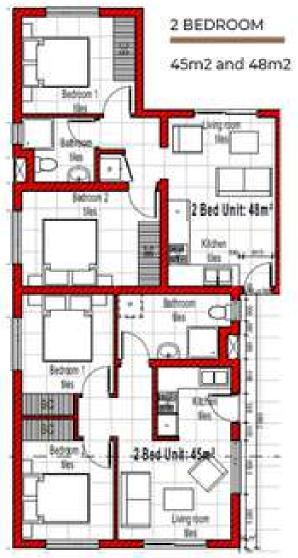
08

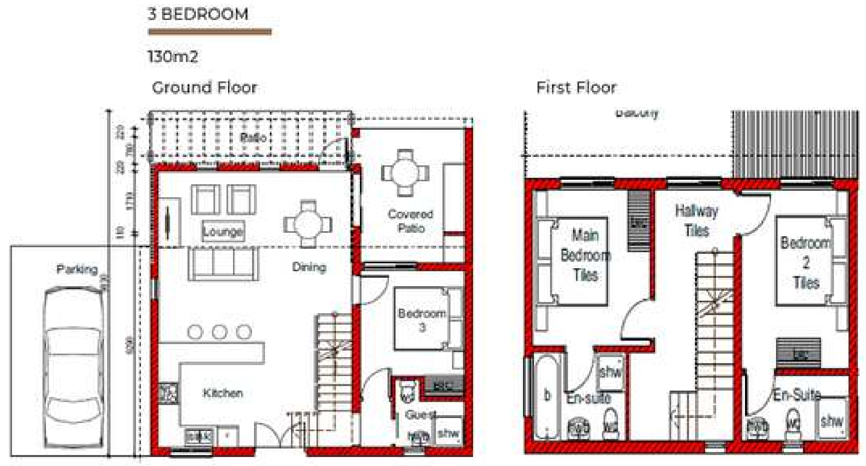
Practice Work
Vheneka Residential Estate

Project Overview

Vheneka Estate is a new residential development in Glendale, Zimbabwe, addressing the growing demand for quality housing following the return of ex-pats after the political transition.

The estate is designed to provide a well-planned, sustainable community with a mix of residential, commercial, and social amenities.





PHASES

LAND USE SCHEDULE : Phase 1

	LAND USES	No.
	LOW-DENSITY RESIDENTIAL	22
1	Rivertree	22 STANDS
	MEDIUM-DENSITY RESIDENTIAL	62
2	Bushwillow	26 TOWNHOUSES
2	Knob Thorn	36 TOWNHOUSES
	HIGH-DENSITY RESIDENTIAL	492
3	Jackal-berry Ridge	276 UNITS
3	Boabab	216 UNITS
4	COMMERCIAL	2
	Retail Centre	1
	Filling Station	1
	BLOCKS	7 - 3 BEDS
		34 - 2 BEDS

LAND USE SCHEDULE : Phase 2

	LAND USES	No.
	HIGH-DENSITY RESIDENTIAL	1440
3	Blackwood Ridge	336 UNITS
3	Marula Tree	648 UNITS
3	Buffalo-thorn	456 UNITS
4	COMMERCIAL	2
	Lodge	1
	Retail Centre	1
5	INSTITUTIONAL	1
	Church & Creche	1
6	OPEN SPACE	1
	BLOCKS	24 - 3 BEDS
		96 - 2 BEDS

thank

YOU